LESSON PLAN

PYRO TECHNIQUES (34)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Warhead system, types and requirements of warhead
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Warhead system, types and requirements of warhead
5. Specific Objectives	34.1.1.1 Trainee will be able to explain various terms on guided weapon.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of warhead terminology diagrams; Instructor supervision required for safety protocols regarding explosive concepts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a warhead in a guided weapon?
(b) Define the term 'fuzing' in the context of a warhead system.
(c) List three common types of warhead effects used in air-to-ground munitions.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and function of warhead system • Types of warheads: HE, HEAT, and Submunition • Fuzing mechanisms: Proximity, Impact, and Delay • Safety requirements for warhead handling • Compatibility with guidance systems	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of High Explosive (HE) warheads • Operational principles of High Explosive Anti-Tank (HEAT) warheads • Characteristics and deployment of cluster/submunition warheads • Selection criteria based on target type and range • Regulatory standards for warhead performance	L & Ds	PPT, PDF	20 min
(c)	• Review of specific warhead terminology • Application of warhead requirements in mission planning • Safety protocols during maintenance and inspection • Troubleshooting common warhead system failures • Summary of key terms and definitions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a proximity fuze and an impact fuze.
(b) Describe the specific requirements a warhead must meet to be effective against armored vehicles.
(c) Identify two safety requirements that must be followed when handling warhead components.

12. Home Assignment and Follow Up	Prepare a brief note comparing the advantages and disadvantages of different propulsion systems, to be discussed in the next lesson on Propulsion system requirements.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Propulsion system, requirements, selection criteria and advantages /disadvantages of each propulsion system	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min